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Global prevalence of Wilms' tumor 1 gene expression and mutations in acute leukemia and myelodysplastic syndromes: a systematic review and meta-analysis

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Abstract

Background: The Wilms Tumor 1 gene (*WT1*) is commonly overexpressed in leukemia, with a magnitude ranging from 28% to 94.5%. Mutation in *WT1* is found in 19% to 68.6% of patients with acute leukemia and myelodysplastic syndrome (MDS). Despite numerous individual studies, the evidence regarding the pooled prevalence of WT1 expression and mutation remains limited.

Objectives: This study aimed to determine the pooled prevalence of *WT1* gene expression and mutations in patients with acute leukemia and MDS, and to evaluate variations across subtypes.

Methods: To conduct this systematic review and meta-analysis, we followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA). The review protocol was registered on PROSPERO (CRD 420251274348), and relevant information was retrieved through comprehensive and reproducible electronic searching of databases, such as MEDLINE, HINARI, Embase, Scopus, CINAHL, the scientific information database, and African online archives. All studies on leukemia and myelodysplastic syndrome from 1997 to January 1, 2026, with data on the prevalence of gene expression and mutation were included, and then inclusion criteria were applied to all relevant articles. Using STATA software (Ver. 11.1), data were analyzed with a random effect model. I^2 test statistics were employed to assess the degree of heterogeneity. The Egger regression test and funnel plot analysis were employed to identify publication bias.

Results: The study included 14,383 participants from 66 articles. The pooled global prevalence of WT1 expression among patients with acute leukemia and MDS was 69.74% (95% CI 63.42–76.07) with the I² statistic ($I^2 = 93.7\%$, $P < 0.001$). In subgroup analyses among AML, MDS, and ALL, respectively, the prevalence was 66.31% (95% CI 51.43–81.19), 66.70% (95% CI 52.50–80.90), and 73.95% (95% CI 66.68–81.22). The prevalence of WT1 mutation among patients with acute leukemia was 15.35% (95% CI 13.06–17.6), and subgroup analysis by type of leukemia among AML and ALL, respectively, was 18.9% (95% CI 9.4–28.4) and 15.35% (95% CI 12.30–16.70).

Conclusions and recommendation: *WT1* overexpression is commonly reported among patients with acute leukemia and MDS, affecting more approximately two-thirds of patients, while WT1 mutation in acute leukemia demonstrates a considerable prevalence. Therefore, assessment of WT1 expression and mutation should be considered in clinical practice to support diagnosis and prognostic evaluation in patients with acute leukemia and MDS.

Keywords: Wilms Tumor 1 (*WT1*), Acute leukemia, Myelodysplastic syndrome, Gene expression, Gene mutation, Meta-analysis

Introduction

The Wilms tumor 1 (*WT1*) gene is located on chromosome 11p13, spanning 50 kilobases, and encodes transcriptional regulatory proteins (1, 2). It is a transcription factor and/or post-transcriptional protein that plays a crucial role in normal hematopoiesis and oncogenesis due to its multifunctional activities in gene regulation, transcription, ribonucleic acid (RNA) processing, proliferation, cell differentiation, survival, and apoptosis (3). In normal hematopoiesis, *WT1* expression is temporary and limited to early progenitor cells, because proper WT1 down-regulation is required for normal lineage maturation. Among *WT1* genes, the proteins lacking the lysine, threonine, and serine insertion function largely as deoxyribonucleic acid (DNA)-binding transcription factors, whereas *WT1* proteins with the insertion have RNA-binding properties and are involved in post-transcriptional regulation (4).

WT1 contributes to the pathogenesis of leukemia and myelodysplastic syndromes (MDS) through both abnormal expression and mutation, as the persistent expression disrupts differentiation and promotes the expansion of immature cells (3). It is a candidate gene for Wilms tumor susceptibility and is highly expressed across hematological malignancies, with predominant expression in acute leukemias, MDS, and chronic myeloid leukemia (CML) with blast crisis. Its overexpression and mutation cause dysregulated hematopoiesis, supporting WT1's position as a major biomarker in leukemia and MDS. In leukemia and MDS, inhibition of differentiation and expansion of immature blast cells result in significant *WT1* overexpression, reflecting leukemic load and disease activity, which promotes proliferation, limits differentiation, and increases malignant blast survival (2, 5). *WT1* expression is observed in 28% to 94.5% of patients with various types of leukemia. Overexpression of wild-type *WT1* and truncated *WT1* isoforms has been found in acute leukemic blood. This trend is also seen in CML, where WT1 overexpression is linked to disease relapse or minimal residual disease (MRD) (6, 7). Interestingly, overexpression of wild type *WT1* and truncated *WT1* was detected in 94.5% and 19% of AML patients, respectively. The truncated form of WT1 is generated by a cryptic translational start site in intron five, resulting in a protein that consists of the C-terminal domain encoded by exons 6–10 only (7, 8).

WT1 overexpression and mutation contribute to leukemic transformation and serve as crucial molecular markers for disease monitoring, burden, and the detection of MRD. Extensive evidence supports the oncogenic role of *WT1* in leukemia, and *WT1*-antisense oligonucleotides have been shown to induce apoptosis in myeloid cell lines (9). WT1 transcript levels at diagnosis and their role in predicting patient outcomes are still a topic of debate. Recently, *WT1* has been recognized as a valuable molecular marker for ALL and AML and is also used to generate *WT1*-specific cytotoxic T-lymphocytes directed against leukemia cells (10, 11).

Despite the growing use of *WT1* expression and mutation as biomarkers in leukemia and MDS, their global prevalence and population-specific variability remain uncertain due to heterogeneity in study design, diagnostic methods, and reporting standards (12). The distribution and clinical relevance of *WT1* alterations across leukemia subtypes, age groups, and regions remain incompletely characterized. *WT1* mutations, which arise during clonal evolution and disrupt regulatory function, contribute to disease progression and relapse, with variable prevalence and impact across populations (13).

Although researchers have studied the molecular mechanisms of WT1-driven leukemogenesis in detail, further exploration of WT1-related signaling pathways in specific leukemia types may help in creating targeted therapies. This could result in better treatment options and improved patient outcomes (14). Therefore, a comprehensive, systematic analysis is needed to clarify the pooled prevalence and clinical significance of *WT1* expression and mutation, thereby improving diagnosis, risk stratification, and disease monitoring (13). This study aimed to determine the pooled prevalence of WT1 gene expression and mutations in patients with acute leukemia and myelodysplastic syndromes, and to evaluate differences across disease subtypes.

Methods

Study design and protocol registration

The main objective of this systematic review and meta-analysis was to determine the global prevalence of *WT1* gene expression and mutations in patients with leukemia. Both published and unpublished studies were conducted to determine the pooled prevalence of *WT1* expression and mutation across various types of hematological malignancies, such as acute leukemias and MDS. The review adhered to the Preferred Reporting Items for Systematic Review and Meta-analysis Protocols (PRISMA-P) guidelines (15). To ensure scientific rigor, the protocol was subsequently registered with PROSPERO with the registration number of CRD 420251274348.

Data source and search strategy

The studies included in this systematic review and meta-analysis were identified through comprehensive, systematic, and reproducible searches of electronic databases such as MEDLINE (PubMed), HINARI, Embase, Scopus, CINAHL, the Scientific Information Database (SID), and African online archives covering publications between 1997 and January, 2026. The gray literature was retrieved from Google Scholar using Publish or Perish. We have also employed manual searching and direct Google searches of the reference lists of already-identified relevant papers to identify additional eligible studies, and the study screening was managed using Rayyan. A comprehensive search strategy has been employed using the condition, context, population, and outcome of interest (Coco Pop) framework. It contained both Medical Subject Headings (MeSH) and free-text phrases. The search included both MeSH terms and free-text keywords related to WT1 gene expression, WT1 mutation, acute leukemia (AML, ALL), myelodysplastic syndrome (MDS), and prevalence, combined with Boolean operators “AND” and "OR." Only English-language studies were included due to the lack of access to full-text non-English publications. The core search strategy included ("Wilms Tumor 1 Protein"[Mesh] OR WT1 OR "Wilms Tumor 1 gene”) AND ("Leukemia, Acute"[Mesh] OR "Leukemia, Myeloid, Myeloblastic, Myelogenous, Acute"[Mesh] OR "Leukemia, Lymphoid, Lymphoblastic, Acute"[Mesh] OR "acute leukemia" OR AML OR ALL) AND ("Myelodysplastic Syndromes"[Mesh] OR MDS OR "myelodysplastic syndrome") AND ("Gene Expression"[Mesh] OR "gene expression" OR "WT1 expression") AND

("Mutation"[Mesh] OR "Genetic Variation"[Mesh] OR "WT1 mutation") AND ("Prevalence"[Mesh] OR "Epidemiology"[Mesh] OR prevalence OR incidence OR frequency OR distribution) AND Humans OR Patients [Mesh].

Study selection and quality assessment

Four authors (SSK, LGB, DF, SA, and GM) independently searched for articles using the following key terms: (1) population (leukemia patients (AML and/or ALL) or patients with MDS); (2) outcome (*WT1* gene expression, *WT1* gene mutation); (3) study design (cross-sectional, comparative cross-sectional, observational, or longitudinal study); and (4) location (globally). The retrieved records were first entered with EndNote X7 to remove duplicate records, then imported into Rayyan for further deduplication, again checked for duplication, and the articles identified through Google Scholar were processed using Publish or Perish to remove duplicates.

Using predefined eligibility criteria, four reviewers (DF, LGB, SA, and GM) independently screened the titles, abstracts, and full texts for data extraction. Any disagreements between four independent reviewers (DF, LGB, SA, and TW) were resolved by SSK to reach a consensus. The methodological quality of the included studies was independently assessed by three reviewers (LGB, DF, and SSK) using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist. Full texts were reviewed to confirm whether each study met the selection criteria or if eligibility was uncertain. The evidence certainty for each outcome was assessed using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach. The evidence certainty was initially considered high and downgraded based on limitations identified in these domains. The five assessment domains include risk of bias, inconsistency, indirectness, imprecision and publication bias. Using these domains, the certainty of evidence was classified as high, moderate, low, or very low. The assessment was performed independently by reviewers, and disagreements were resolved through consensus. The methodological validity of each study was evaluated using the Joanna Brigg Institute (JBI) quality assessment manual (16). This tool was used thoroughly to evaluate aspects, such as sample representativeness, participant recruitment, sample size, measurement reliability, and appropriateness of statistical analysis. Each item was rated as “Yes,” “No,” or "Unclear." Studies with poor quality were rated as less than 50% (15), and high-quality studies received ratings higher than 50% (15). The final analysis included studies that satisfied at least 50% of the quality criteria, which were deemed to have acceptable methodological quality. Reviewers used discussion and consensus to settle any disputes.

Eligibility criteria

The content of each article was independently and meticulously appraised by five authors (TW, SA, TW, LG, and DF). Studies were considered eligible for inclusion if they met the following criteria: institutionally based investigations of WT1 gene expression and mutation between 1997 and January, 2026. The selection process involved evaluating titles and abstracts, followed by a review of full-text papers. Case reports, articles lacking full texts or abstracts, and studies that did not report the outcomes of interest were excluded. The identification, screening, inclusion, and exclusion of records were conducted in accordance with the PRISMA flow diagram guidelines (15). After thoroughly reviewing the abstracts and full texts, five reviewers (TW, LGB, SA, GM, and DF) selected eligible studies. Studies with methodological issues were excluded. Any disagreements during the review process were resolved through discussion and consensus. When consensus could not be reached, a third reviewer (SSK) was consulted to make the final decision.

Population: Studies carried out on patients (both pediatric and adult patients) with acute leukemia and myelodysplastic syndromes

Study area: Studies conducted worldwide and published between 1997 and January, 2026, were included.

Study design: Original studies and institutional-based surveys or cross-sectional studies, comparative cross-sectional, observational, longitudinal or cohort (retrospective and prospective) studies, experimental studies, and clinical trials documenting *WT1* gene mutations and expression patterns in acute leukemia (AML and ALL) and MDS worldwide were considered.

Language: Only studies published in the English language were included.

Publication condition: Research that met the criteria for eligibility was included, regardless of whether it was published, unpublished, gray literature, etc. Research on the magnitude of WT1 gene mutations and expression patterns in acute leukemia and MDS which was published worldwide between 1997 and January 2026.

Diagnostic and detection methods: The diagnosis of acute leukemia (AML and ALL) and MDS was established using standard hematological diagnostic criteria, including peripheral blood and bone marrow morphology, cytochemical staining, and cytogenetic analysis. WT1 gene expression or mutation was primarily detected using quantitative polymerase chain reaction (qPCR) and was considered in the study.

Data extraction and abstraction

The five authors validated the required data using a pilot-tested data abstraction tool. The retrieved data included the first author, study area, study design, publication year, sample size, response rate, mean *WT1* level, mean *WT1* mutation percent, and the prevalence of *WT1* mutation among patients with acute leukemia and MDS. Any disagreements that arose during the data extraction process were resolved through discussion and, if necessary, with the involvement of a fifth person. The quality of the studies was assessed by the JBI procedure (15), and the assessment results were determined by taking the average score of the five authors. After evaluating the methodological quality and overall characteristics of the publications, data were entered into a data extraction sheet using Microsoft Excel. For each article that met the eligibility criteria, we extracted the first author's name, total sample size, publication year, study design, study area, number of cases, and proportion, which were entered into a data extraction sheet using Microsoft Excel. For each article that met the eligibility criteria, we extracted the first author's name, total sample size, publication year, study design, study area, number of cases, and proportion.

Outcome of interest: The outcome of interest was the pooled prevalence of *WT1* expression and gene mutation patterns among patients with acute leukemia and MDS and its variation with leukemia subtypes.

Publication bias and heterogeneity

A funnel plot was used to visually inspect the impact of small study value effects or publication bias on the estimated pooled proportion. The Egger's test indicated a strongly significant publication bias and subgroup analysis was done by age category, WHO region, country status and year of the study. Finally, sensitivity analysis was performed to assess the robustness of the synthesized results on WT1 gene expression and mutation.

Data analysis

Data were initially extracted using Microsoft Excel and subsequently imported into STATA software version 11 for advanced statistical analysis. The general characteristics of the included studies, such as the first author's name, year of publication, study area and setting, study design, and sample size, were summarized in a tabular format.

The pooled prevalence of *WT1* mutation and *WT1* gene expression was calculated based on the reported prevalence and standard error from each original study. Forest plots were generated to visually represent the pooled prevalence estimates. Heterogeneity among studies was assessed using the I^2 statistic, with substantial heterogeneity defined as $I^2 > 50\%$ and a corresponding p value less than 0.05. Potential publication bias was evaluated using both Egger's and Begg's correlation tests, with statistical significance set at the 5% level.

Results

A total of 5,215 records were found through database searching, which included PubMed (MEDLINE), Google Scholar, ScienceDirect, Embase, and the Cochrane Library, along with manual searches. After removing 2,446 duplicate records, 2,769 records were left for screening. Of these, automation tools excluded 1,871 records, and title and abstract screening removed 707 records. Then, 191 reports were requested for retrieval, but 76 were not retrieved. A total of 115 full-text articles were evaluated for eligibility, and 52 were excluded for various reasons: missing data (n = 31), not meeting the inclusion criteria (n = 8), involving patients without leukemia or myelodysplastic syndrome (n = 4), and other reasons (n = 9). In the end, 63 studies made up of 66 reports were included in the final systematic review and meta-analysis (Fig. 1).

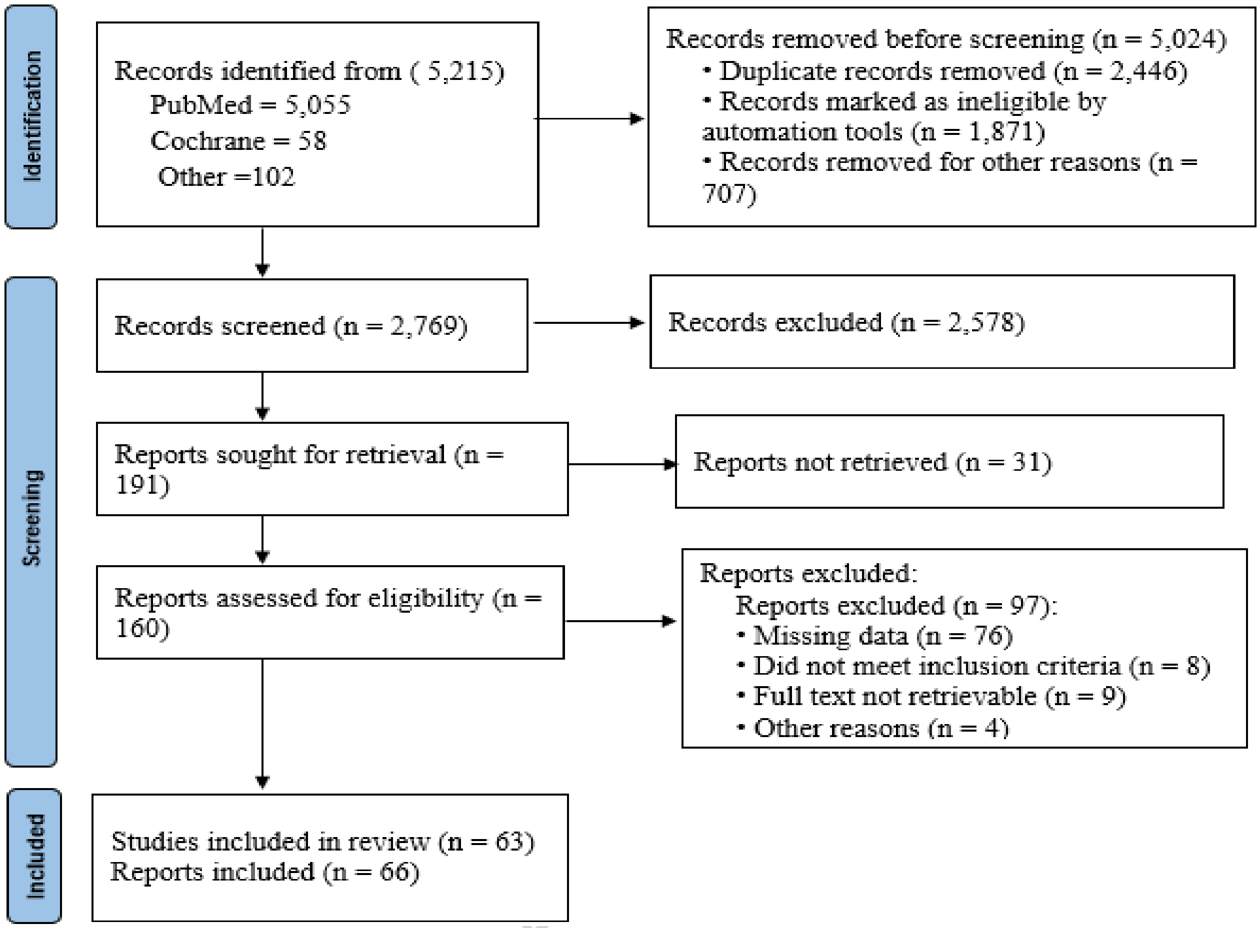

Figure 1 PRISMA flow chart of studies selected for the systematic review and meta-analysis

Description of the studies

The present study included 14,383 participants derived from 66 articles published between 1997 and January, 2026. The systematic review and meta-analysis incorporated studies conducted across multiple countries, including nine from China, eleven from the United States, ten from Germany, seven from Egypt, four from Iran, four from the United Kingdom, three from India, two each from Italy, France, Austria, and Taiwan, and one each from Turkey, Hungary, Israel, Poland, Japan, the Netherlands, and Thailand (Table 1).

Table 1 Summary of study articles included in the global prevalence of *WT1* gene expression systematic review and meta-analysis

Authors name	Year of publication	Study area	ountry statu	Study design	ge categor	lignancy t	al sample	No of cas	expression	oportion(	log prev	SE. Prop	LB	UB
Ahmed El et al	2020	Egypt	Developing	Cross sectional	Adult	AML	42	31	positive	73.8	4.30135873	6.785068	60.5013	87.09873257
Cilloni D et al	2003	Italy	Developed	Cross sectional	Adult	MDS	173	131	positive	78	4.35670883	3.149456	71.8271	84.1729344
Rennevile A et al	2009	France	Developed	Clinical trail	Pediatrics	ALL	146	123	positive	84	4.4308168	3.034053	78.0533	89.94674448
Abd-Ekodobusa R	2018	Egypt	Developing	Cross sectional	Both	ALL	30	21	positive	70	4.24849524	8.3666	53.6015	86.39853652
Ozgan Unsal et al	2000	Turkey	Developing	Cross sectional	Pediatrics	ALL	28	22	positive	78.5	4.36309862	7.763813	63.2829	93.71707265
Routenberg C et a	2016	Germany	Developed	Cross sectional	Adult	AML	60	40	positive	66.6	4.19870458	6.088842	54.6659	78.53413082
Routenberg C et a	2016	Germany	Developed	Cross sectional	Adult	MDS	43	19	positive	44.4	3.79323947	7.576954	29.5492	59.25082992
Yin M et al	2020	China	Developing	Cohort	Pediatrics	ALL	188	147	positive	78.31	4.36067531	3.005796	72.4186	84.20135963
Ayatollah H et al	2017	Iran	Developing	Cross sectional	Both	ALL	126	104	positive	81	4.39444915	3.494894	74.15	87.8499927
Bergmann L et al	1997	Germany	Developed	Cross sectional	Adult	AML	161	24	positive	77	4.34380542	3.316625	70.4994	83.50058459
Busse A et al	2009	Germany	Developed	Cross sectional	All	ALL	219	238	positive	92	4.52178858	1.83323	88.4069	95.59312992
Gaiger A et al	1999	Austria	Developed	Cohort	Pediatrics	ALL	50	30	positive	60	4.09434456	6.928203	46.4207	73.57927833
Brieger J et al	1995	Germany	Developed	Cross sectional	Both	AML	83	67	positive	83	4.41884061	4.123106	74.9187	91.08128703
Zsófia Ujj et al	2016	Hungary	Developed	Cohort	Adult	AML	53	60	positive	88.3	4.48074011	4.415047	79.6465	96.95349245
Wang S et al	2020	China	Developing	Cross sectional	Both	ALL	162	107	positive	66.6	4.19870458	3.705551	59.3371	73.86288073
M Weisser, et al	2005	Germany	Developed	Cohort	Both	AML	44	16	positive	36.36	3.59346927	7.251881	22.1463	50.57368727
H Shima et al	2012	China	Developing	Cohort	Both	MDS	147	121	positive	82.3	4.41037111	3.14795	76.13	88.46998282
Heesch S et al	2010	Germany	Dcvelepcd	Cross sectional	Adult	ALL	252	218	positive	86.5	4.46014441	2.152656	82.2808	90.71920609
Lyu X et al	2014	USA	Developed	Cohort	Adult	AML	103	29	negative	28	3.33220451	4.424117	19.3287	36.67127027
Asgarian H et al	2005	Iran	Developing	Cohort	Adult	AML	11	10	positive	91	4.51085951	8.628705	74.0877	107.9122613
Rautenberg, C e	2019	Germany	Developed	Cross sectional	Adult	MDS	94	54	positive	57	4.04305127	5.106316	46.9916	67.00838032
Guo X et al	1999	China	Developing	Cohort	Pediatrics	AML	68	36	positive	53	3.97029191	6.052467	41.1372	64.8628347
Lim CK et al	2007	China	Developing	Cross sectional	Adult	ALL	18	3	positive	16.7	2.81540872	8.791126	-0.5306	33.93060613
Lim CK et al	2023	China	Developing	Cross sectional	Adult	ALL	18	15	positive	83.3	4.42244855	8.791126	66.0694	100.5306061
M YIN, et al	2023	China	Developing	Cross sectional	Pediatrics	ALL	183	130	positive	71.04	4.2632431	3.352936	64.4682	77.61175375

Log. Prev: logarithm of prevalence, SE. Prop: standard error of the proportion, LB: lower bound of confidence interval, UB: upper bound of confidence interval

Table 2 Summary of included articles for the study of the global prevalence of *WT1* gene mutation among acute leukemia patients, systematic review, and meta-analysis

Authors name	Year of publication	**Study area**	**Country status**	**Study design**	**leukemia type**	**age category**	**Total ample**	**No of case**	**Proportion(%)**	**log prev**	**SE. Prop**	**LB**	**UB**
Hollink IH et al	2008	Egypt	Developing	Cross sectional	ALL	Adult	298	35	12	2.484907	1.8824516	8.310395	15.68960521
Ma W et al	2009	USA	Developed	Cross sectional	AML	Adult	174	12	6.9	1.931521	1.9214309	3.133996	10.66600449
Heesch S et al	2010	Germany	Developed	Cross sectional	ALL	Adult	238	20	8	2.079442	1.7585326	4.553276	11.44672393
G Toogeh et al	2015	Iran	Developing	Cross sectional	AML	Adult	88	11	12.5	2.525729	3.5254755	5.590068	19.40993192
Feng-Ming Tien et al	2018	Taiwan	Developed	Cross sectional	AML	Adult	766	90	11.8	2.4681	1.1656305	9.515364	14.08463583
Naghmeh Niktoreh et a	2019	Germany	Developed	Cross sectional	AML	Adult	353	125	35.53	3.570377	2.5473527	30.53719	40.52281136
Jing Xu et al	2020	China	Developing	Cross sectional	AML	Adult	220	28	12.7	2.541602	2.2449033	8.299989	17.1000105
Long Su et al	2018	China	Developing	Cross sectional	AML	Adult	81	15	18.52	2.918851	4.3162192	10.06021	26.97978962
Yin Wang et al	2021	China	Developing	Cross sectional	AML	Adult	870	58	6.7	1.902108	0.847654	5.038598	8.361401768
Salehzadeh S et al	2019	Italy	Developed	Cross sectional	AML	Adult	98	13	12.9	2.557227	3.3860307	6.26338	19.53662022
Ho PA et al	2010	USA	Developed	Cross sectional	AML	Pedtriacs	842	70	8.3	2.116256	0.9507529	6.436524	10.16347572
S Aref et al	2014	Egypt	Developing	Cross sectional	AML	Both	82	11	13.41	2.596001	3.7630616	6.034399	20.78560082
A Nagler etal	2025	Israel	Developed	Cross sectional	AML	Adult	703	50	7.11	1.961502	0.969263	5.210245	9.009755413
L. King-Underwood et	1996	UK	Developed	Cross sectional	ALL	Adult	34	4	11.11	2.407846	5.3894447	0.546688	21.67331171
L. King-Underwood et	1996	UK	Developed	Cross sectional	AML	Adult	34	5	14	2.639057	5.9507785	2.336474	25.66352585
HA Hou et al	2010	Taiwan	Developed	Cross sectional	AML	Both	470	32	6.8	1.916923	1.1612173	4.524014	9.075985836
Anitha, G.R.J. et al	2024	India	Developing	Cross sectional	AML	Adult	150	19	12.6	2.533697	2.7095387	7.289304	17.91069586
H Atluri, et al	2023	USA	Developed	Cross sectional	AML	Adult	161	84	52	3.951244	3.9373983	44.2827	59.71730072
D Koczkodaj,et al	2022	Poland	Developed	Cross sectional	AML	Adult	90	26	28.9	3.363842	4.7781796	19.53477	38.26523196
V Gannamani et al	2024	USA	Developed	Cross sectional	AML	Adult	53	18	34	3.526361	6.5068904	21.24649	46.75350518
PR Ugale et al	2024	India	Developing	Cross sectional	AML	Adult	506	68	13.44	2.598235	1.5162925	10.46807	16.41193339
Miyagawa K et al	1999	Japan	Developed	Cross sectional	AML	Adult	46	6	13	2.564949	4.9585236	3.281294	22.7187063
Salah Aref et al	2013	Egypt	Developing	Cross sectional	AML	Both	82	11	13.41	2.596001	3.7630616	6.034399	20.78560082
Yu, T. et al	2023	China	Developing	Cohort	AML	Adult	269	153	56.88	4.040944	3.0195556	50.96167	62.79832896
P Virappane et al	2008	UK	Developed	Cross sectional	AML	Adult	470	47	10	2.302585	1.3837968	7.287758	12.71224175
A Renneville et al	2009	France	Developed	Cross sectional	ALL	Pedtriacs	146	15	10	2.302585	2.4828177	5.133677	14.86632262
NL Mikhael et al	2019	Egypt	Developing	Cross sectional	ALL	Pedtriacs	140	96	68.6	4.228293	3.9224992	60.9119	76.28809844
IHIM Hollink et al	2009	Netherlands	Developed	Cross sectional	AML	Pedtriacs	298	35	12	2.484907	1.8824516	8.310395	15.68960521
I Haider et al	2020	India	Developing	Cross sectional	AML	Adult	100	39	**39**	3.663562	4.8774994	29.4401	48.55989874
Zidan et al	2014	Egypt	Developing	Cross sectional	AML	Adult	216	23	10.65	2.36556	2.0989167	6.536123	14.76387677
Y Wang et al	2021	USA	Developed	Cross sectional	AML	Adult	870	58	6.7	1.902108	1.2775874	4.195929	9.204071343
H Awada et al	2020	*USA*	*Developed*	Cohort	AML	Adult	2188	109	5	1.609438	0.4659327	4.086772	5.913227997
Lauhakirti D et al	2011	Thailand	Developing	Cross sectional	AML	Adult	400	49	12.24	2.504709	1.6387361	9.028077	15.45192275
Paschka P et al	2008	USA	Developed	Cross sectional	AML	Adult	196	21	10.7	2.370244	2.2079518	6.372415	15.02758547
Marivel J et al	2016	UK	Developed	Cross sectional	AML	Adult	50	3	6	1.791759	3.3585711	-0.5828	12.5827994
Rostami S et al	2021	Iran	Developing	Cohort	AML	Adult	130	7	5.4	1.686399	1.6910004	2.085639	12.27500041
M-T Krauth et al	2014	Austria	Developed	Cross sectional	AML	Adult	3157	175	5.5	1.704748	0.4057513	4.704727	6.295272618
VI Gaidzik et al	2009	Germany	Developed	Cross sectional	AML	Adult	617	78	12.6	2.533697	1.3359752	9.981489	15.21851146
Becker H et al	2010	USA	Developed	Cross sectional	AML	Adult	243	17	7	1.94591	1.6367689	3.791933	10.20806699
V Tosello et al	2009	USA	Developed	Cross sectional	ALL	Adult	85	10	11.77	2.465554	1.6367689	8.561933	14.97806699
V Tosello et al	2009	USA	Developed	Cross sectional	ALL	Pedtriacs	211	28	13.2	2.580217	1.6367689	9.991933	16.40806699

Log. Prev: logarithm of prevalence, SE. Prop: standard error of the proportion, LB: lower bound of confidence interval, UB: upper bound of confidence interval

Meta-analysis revealed a higher pooled prevalence of *WT1* gene expression

In this meta-analysis, a forest plot was used to estimate the pooled prevalence and to present the effect size of each included study, along with their corresponding 95% CIs (Fig. 2). Based on 25 eligible studies, the pooled global prevalence of *WT1* expression among patients with acute leukemia and MDS was 69.74% (95% CI 63.42–76.07%). A high level of heterogeneity was observed among the included studies, as indicated by the I^2 statistic ($I^2 = 93.7\%$, $P < 0.001$). The funnel plot showed asymmetry, suggesting the presence of small-study effects and potential publication bias (Supplementary Figure 1). Publication bias was further assessed using Egger's regression test, which demonstrated statistically significant asymmetry ($P < 0.001$; Supplementary Table 1). Both the funnel plot and Egger's test were applied to evaluate publication bias, with statistical significance set at $P < 0.05$. Consequently, a trim-and-fill analysis was performed to adjust for potential publication bias (Supplementary Table 2). In addition, a sensitivity analysis was conducted to assess the influence of individual studies on the overall pooled estimate (Supplementary Figure 4). The prevalence estimates reported by the individual studies ranged from 16.7% to 92%, with corresponding variations in the width of their CIs. The pooled estimate and its 95% CI are represented by the diamond at the bottom of the plot, while the individual study estimates and their respective confidence intervals are shown by the squares and horizontal lines.

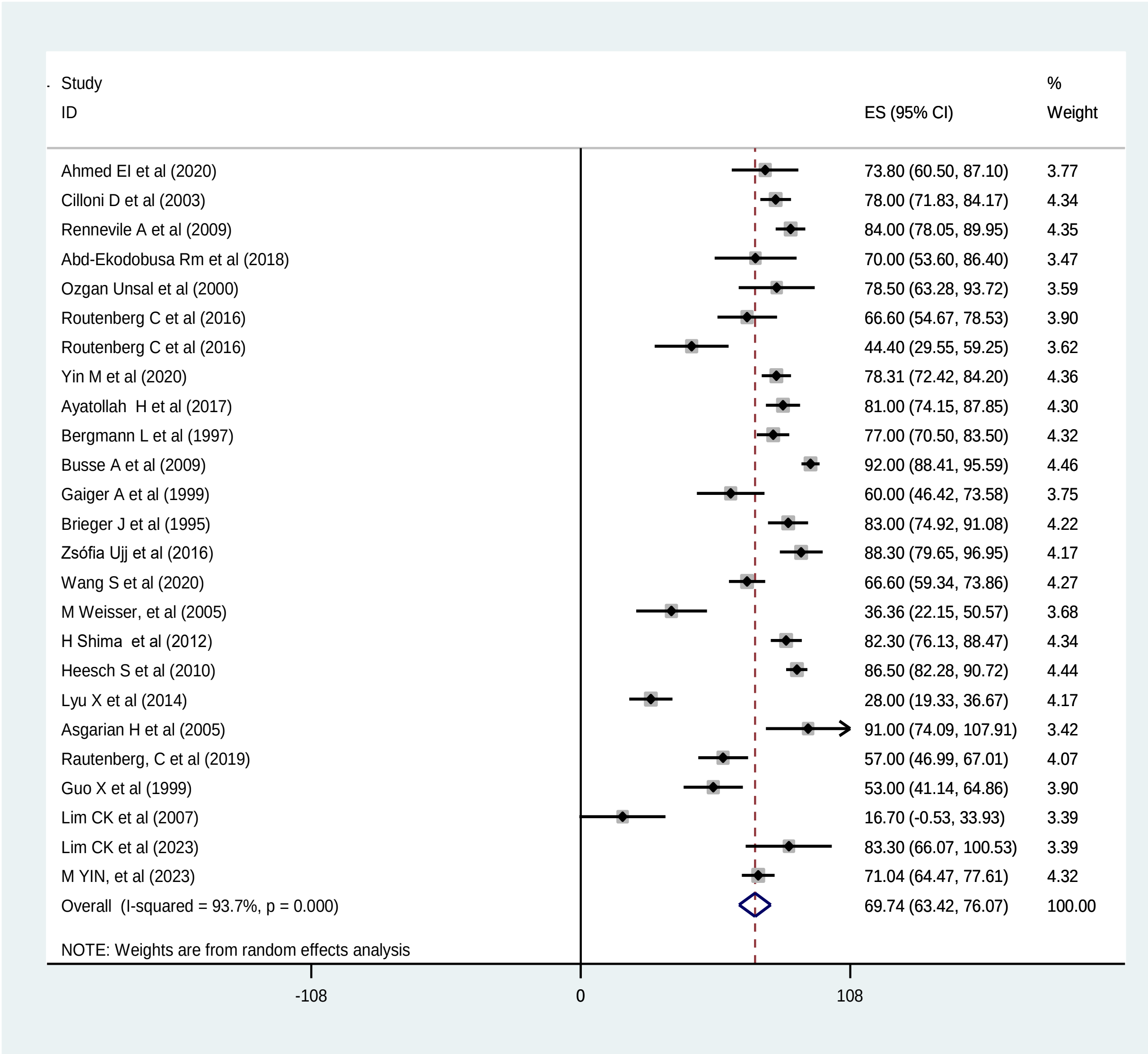

Figure 2 Forest plot for pooled prevalence of *WT1* gene expression among acute leukemia and MDS patients

The vertical dashed line corresponds to the overall pooled prevalence estimate of approximately 69.7%. Studies located to the left of this line report lower than the average *WT1* expression, whereas those positioned to the right indicate higher-than-average prevalence. The relatively symmetric distribution of studies around this line suggests that no single group of studies consistently drives the pooled estimate; however, several outlying estimates may contribute to the overall variability observed across the included studies.

Subgroup analysis

To reduce the random variance between the point estimates of the primary study, we performed subgroup analysis by type of hematological malignancy, age category, and country status, and then we adjusted the observed variability. Because all studies assessed *WT1* gene expression or mutation using the same detection method (qPCR), subgroup analyses based on detection method were not performed.

***WT1* gene expression is prevalent across different types of hematological malignancies**

The subgroup analysis comprised 25 studies. The findings revealed that the prevalence of *WT1* expression in AML, MDS, and ALL was 66.31% (95% CI 51.43, 81.19), 66.70% (52.50, 80.90), and 73.95% (66.68, 81.22), respectively. The substantial heterogeneity was observed among the included studies ($I^2 = 94.9\%$, $p < 0.001$) in AML (Fig. 3).

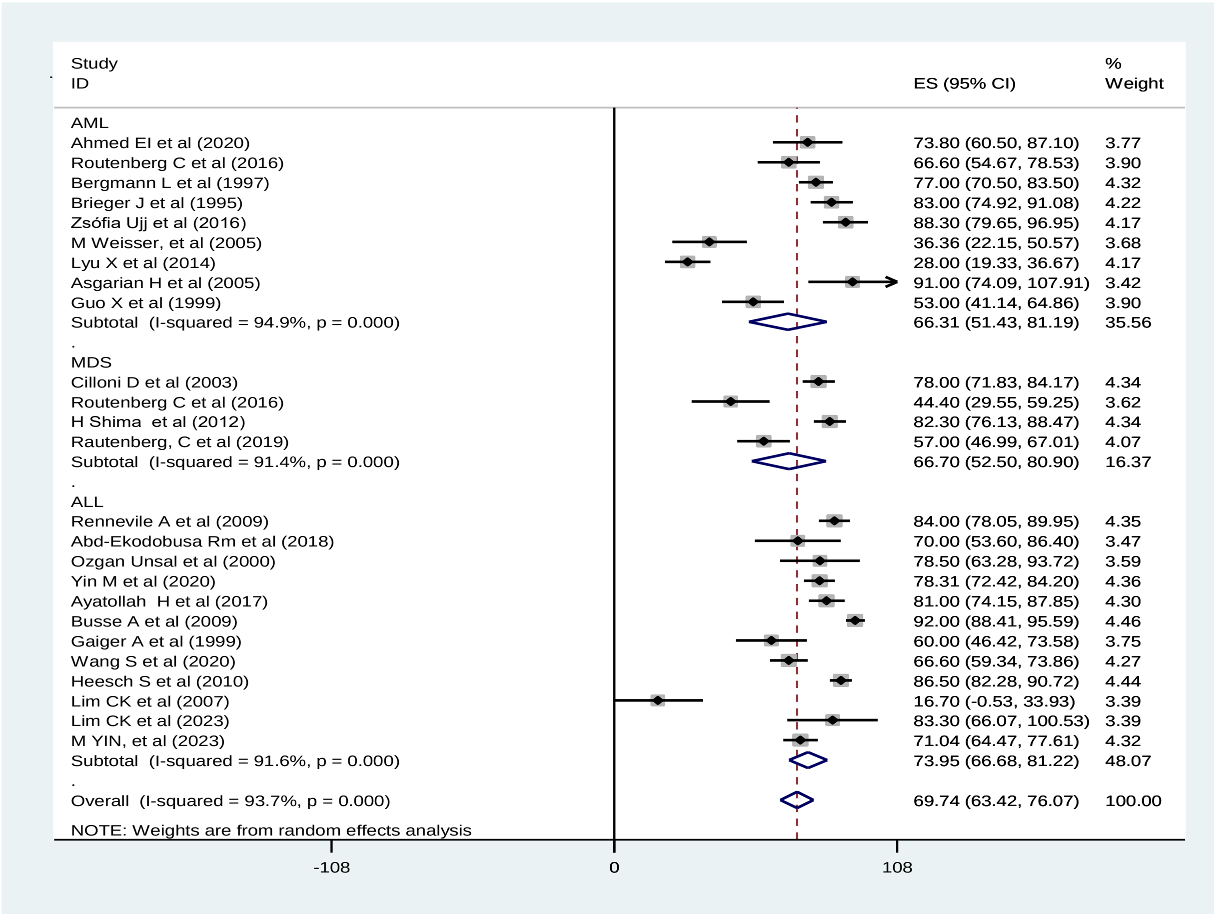

Figure 3. Subgroup analysis of *WT1* expression by type of hematological malignancy

Subgroup analysis of *WT1* expression by age category and country status

This figure presents a subgroup forest plot showing the pooled prevalence of *WT1* overexpression across the included studies, stratified by age category (adult, pediatrics, and both), using a random-effects meta-analysis model. In the adult subgroup, the pooled prevalence of *WT1* overexpression was 66.19% (95% CI 54.30–78.09%), whereas in the pediatric subgroup, the pooled prevalence was 71.67% (95% CI 63.21–80.14%), and for studies that included both adult and pediatric populations, the pooled prevalence was 74.26% (95% CI 63.83–84.68%), with high heterogeneity ($I^2 = 93.2\%$, $p < 0.001$) (Supplementary Figure 2). On the other hand, the results of this systematic review and meta-analysis showed that the prevalence of *WT1* expression varied by country status, with 71.13% reported in developing countries (95% CI 63.72–78.54) and 68.55% in developed countries (95% CI 58.84–78.28) (I^2 = 95.8%, $P < 0.000$) (Supplementary Figure 3).

Sensitivity analysis

Each study had no effect on the pooled estimate, indicating that the results were reliable. Pooling the proportion by removing one trial at a time yielded consistent and reliable results (Supplementary Figure 4).

Meta-analysis of the *WT1* gene mutation

The meta-analysis of *WT1* gene mutations among leukemia patients was conducted using 41 eligible articles. A forest plot (Fig. 5) was used to estimate the pooled prevalence and display the effect size of each included study along with its corresponding 95% CIs. The global pooled prevalence of *WT1* mutation in acute leukemia was 15.35% (95% CI 13.06–17.64%). Substantial heterogeneity was observed among the included studies, as indicated by the I^2 statistic (I^2 = 96%, $P < 0.001$). The funnel plot appeared asymmetrical around the pooled effect, suggesting potential publication bias or small-study effects. Some variability, reflecting underlying heterogeneity, was also observed (Supplementary Figure 5). Assessment of publication bias using Egger's regression test revealed statistically significant bias ($P < 0.001$; Supplementary Table 3). Trim-and-fill analysis was conducted to adjust for potential publication bias (Supplementary Table 4). In addition, a sensitivity analysis was conducted to assess the impact of individual studies on the overall pooled estimate (Fig. 10).

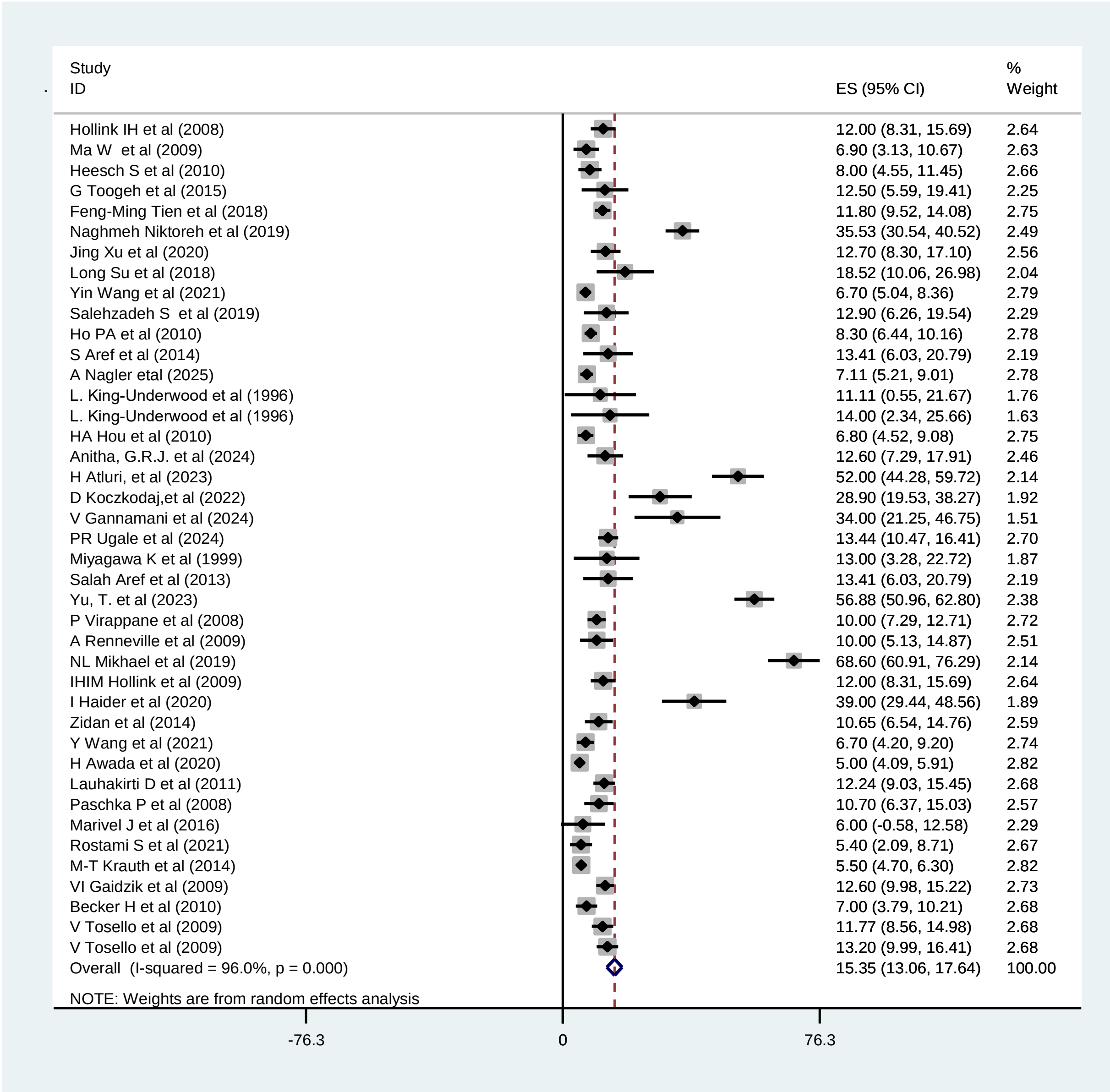

Figure 4 Forest plot for pooled prevalence of *WT1* gene mutation among acute leukemia patients

Subgroup analysis

To minimize random variation between the point estimates of the primary studies, we conducted subgroup analyses based on leukemia type, age category, WHO regional classification, and country status and subsequently adjusted for the observed variability. Since all included studies measured

WT1 gene expression or mutation using quantitative PCR (qPCR), subgroup analyses based on detection method were not conducted.

Subgroup analysis of the *WT1* gene mutation by type of leukemia

The findings of this systematic review and meta-analysis revealed that the prevalence of *WT1* mutation by type of leukemia was reported as 18.91% in ALL (95% CI, 9.39, 28.44) and 15.35% in AML (95% CI, 12.30, 16.69) with high heterogeneity (I^2 = 95.6%, P = 0.000) (Fig. 5).

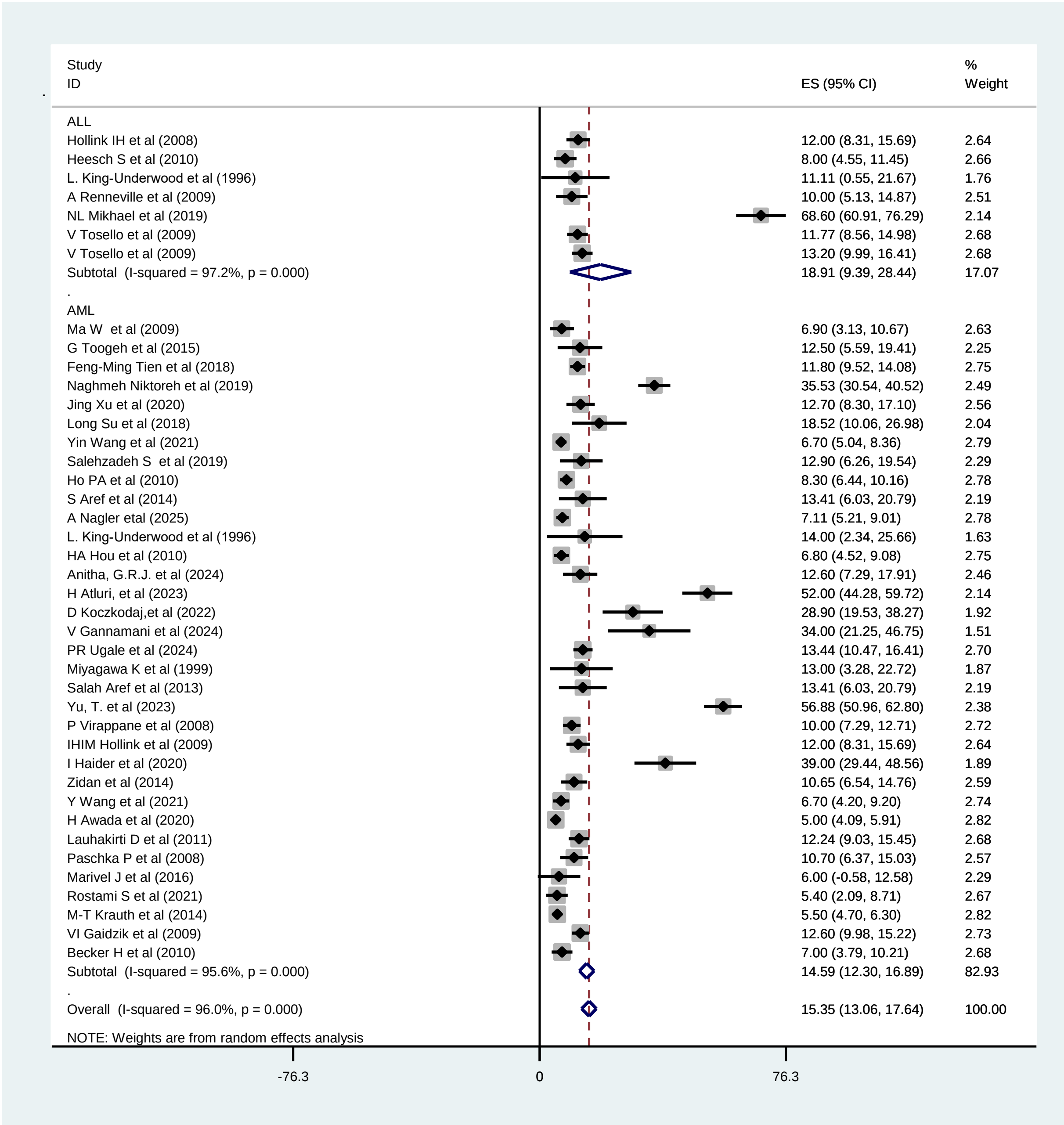

Figure 5 Forest plot for subgroup analysis of *WT1* gene mutation by type of leukemia

Subgroup analysis of the *WT1* gene mutation by age category, country status, study year, and WHO region

The findings of this systematic review and meta-analysis showed that the prevalence of WT1 mutation by age group was 14.83% in adults (95% CI 12.41–17.24) and 21.83% in pediatric patients (95% CI 10.00–33.66) with high heterogeneity (I^2 = 98.2%, P = 0.000) (Supplementary Figure 6).

The evidence synthesized in this systematic review and meta-analysis indicated that the prevalence of *WT1* mutation by country status was 20.18% in developing countries (95% CI 13.66–26.69) and 12.48% in developed countries (95% CI 10.36–17.64) with high heterogeneity (I^2 = 93.8%, P = 0.000) (Supplementary Figure 7).

The outcomes of this systematic review and meta-analysis revealed that the prevalence of *WT1* expression by year of the study was reported as 10.02% in studies before 2016 (95% CI, 8.45, 11.58) and 21.05% in studies during and after 2016 (95% CI, 16.41, 25.70) with heterogeneity (I^2 = 97.8%, P = 0.000) (Supplementary Figure 8).

The synthesized evidence generated by this systematic review and meta-analysis, showed that the prevalence of the *WT1* mutation by the WHO region was reported in the African region, the region of America, the European region, the EMRO (East Mediterranean region), the WPRO (Western Pacific region), and the SEARO (South-East Asia Region) as 23.45% (95% CI 6.47, 40.44), 13.50% (95% CI 9.49, 17.51), 13.52 (95% CI 9.25, 17.79), 7.15 (95% CI 4.63, 9.67), 17.82 (95% CI 9.63, 26.01), and 17.65 (95% CI 10.94, 24.35) with heterogeneity (I^2 = 89.2%, P = 0.000) (95% CI 10.94, 24.36), respectively (Supplementary Figure 9).

Sensitivity analysis

Each study had no effect on the pooled estimate, indicating that the results were reliable. Pooling the proportion by removing one trial at a time yielded consistent and reliable results (Supplementary Figure 10).

Summary of GRADE evidence profile for *WT1* expression and mutation

WT1 expression in AML, based on eight cross-sectional and observational studies, showed a pooled prevalence of 66.31(95% CI 51.43, 81.19) with low certainty due to high heterogeneity and study design; in MDS, four cross-sectional studies reported 66.70 (95% CI 52.50, 80.90) with very low certainty due to serious inconsistency and small sample size (Table 3). The subgroup analysis showed that WT1 gene mutation in ALL, based on seven studies, had a pooled prevalence of 18.91% (95% CI 9.39–28.44), and very high heterogeneity (I^2 = 97.2%), resulting in low certainty of evidence due to variability and observational study design. Similarly, in AML, based on 34 studies, the pooled prevalence was 15.35% (12.30–16.89), and high heterogeneity (I^2 = 95.6%), with the certainty of evidence also rated as low due to very high heterogeneity and the observational nature of the included studies.

Table 3 Summary of findings table presenting pooled estimates and certainty of evidence for WT1 gene expression by leukemia type is provided below based on the GRADE approach

Outcome	No. of studies	Study design	Pooled effect	95% CI	Certainty of	Reasons for rating

			(random effects)		evidence (GRADE)	
WT1 expression in AML	8 studies	Cross-sectional/observational	66.31%	51.43–81.19	Low	High heterogeneity (I^2 = 94.9%) and observational design
WT1 expression in MDS	4 studies	Cross-sectional	66.70%	52.50–80.90	Very low	Inconsistency (I^2 = 91.4%) and small number of studies
WT1 expression in ALL	9 studies	Cross-sectional/observational	73.95%	66.68–81.22	Low	High heterogeneity (I^2 = 91.6%) and moderate risk of bias
Overall pooled estimate	25 studies	Observational	69.74%	63.42–76.07	Low	Very high heterogeneity (I^2 = 93.7%), observational evidence, inconsistency

Discussion

WT1 is commonly expressed in organs involved in the developing hematopoietic system and is largely restricted to early progenitor cells. However, expression gradually diminishes to nearly undetectable levels as cells transition to mature myeloid and lymphoid cells (17). Mutations in

WT1 occur in both syndromes associated with acute leukemia and MDS (18). WT1 overexpression is a hallmark of dysregulated hematopoiesis and increased immature blasts in leukemia and MDS patients and is an indicator of genomic instability and aberrant transcriptional control that underlie malignant transformation. *WT1* overexpression in leukemia and MDS is important in diagnosis by helping to identify malignant hematopoietic blasts and in prognosis by indicating poorer outcomes in patients with higher levels of WT1. It is also used as an indicator of an increased leukemia burden, monitoring MRD, and early prediction of relapse, which is associated with disease severity (19, 20). *WT1* mutations in leukemia and MDS are present in a subset of leukemic cells, often associated with distinct molecular and cytogenetic subgroups that lead to clonal evolution in leukemia and MDS patients. *WT1* mutations provide prognostic information and are frequently observed in leukemia and MDS patients, which is important in diagnosis, prognosis and monitoring leukemia and MDS in various subtypes and age groups (5, 21, 22).

According to this systematic review and meta-analysis, the pooled global prevalence of WT1 expression among patients with acute leukemia and MDS was 69.74% (95% CI 63.42–76.07). This indicates that approximately two-thirds of patients with these hematological malignancies possess *WT1* overexpression. The high prevalence of *WT1* expression may result from dysregulated hematopoiesis associated with malignancy. In healthy individuals, *WT1* expression is tightly controlled and largely restricted to hematopoietic progenitor cells. However, in acute leukemia and MDS, malignant transformation disrupts normal differentiation, leading to the expansion of immature blast cells that inherently overexpress *WT1* (23). The cellular proliferation, genomic instability, and aberrant transcriptional control lead to sustained *WT1* activation as part of the leukemic program. As a result, broad hematopoietic dysregulation and leukemic blast accumulation give a biologically reasonable explanation for the high prevalence of *WT1* overexpression seen in patients with acute leukemia and MDS (24). Clinically, *WT1* overexpression serves multiple roles as a diagnostic biomarker for identifying malignant blasts, as a prognostic indicator, where higher expression predicts poorer outcomes, and as an MRD marker for early detection of relapse (21, 22, 25, 26).

The findings revealed substantial heterogeneity among the included studies ($I^2 = 93.7\%$, $p < 0.000$), indicating considerable variability in the reported prevalence estimates. This high level of heterogeneity may be attributed to differences in study characteristics, including patient populations such as AML, ALL, or MDS; age groups; disease status; laboratory methods used for *WT1* detection (RT-PCR protocol variations); and cutoff thresholds for defining positivity, as well as variations in study design and diagnostic capacity across different settings (27).

The reliability of evidence regarding outcomes was generally rated as low to very low based on the GRADE method. This was mainly attributed to the observational design of the studies included and the presence of very high heterogeneity ($I^2 > 90\%$), which reflects significant variability among different study conditions, populations, and methodologies. While combined estimates indicate a relatively high occurrence of WT1 expression in hematological cancers, the considerable inconsistency and possible bias raise concerns about the validity of these results. Consequently, these findings should be approached with caution, and additional well-structured research is required to enhance the evidence foundation.

The findings showed that the estimated prevalence of *WT1* expression was 66.31% (95% CI 51.43–81.19) in AML, 66.70% (95% CI 52.50–80.90) in MDS, and 73.95% (95% CI 66.68–81.22) in ALL. The overlapping confidence intervals across these conditions suggest that WT1 overexpression represents a common molecular feature of malignant hematopoiesis rather than disease-specific expression. This aligns with its established role as a leukemia-associated marker, serving as a sensitive indicator of malignant transformation and disease burden (28). The high *WT1* expression in patients with AML might be due to the utilization of MRD and is a predictor of poor outcomes, validating the substantial prevalence found in this investigation (22). In the case of MDS, elevated *WT1* expression might be due to increased blast cells that progress toward overt leukemia (29).

The highest pooled prevalence of *WT1* expression was observed in ALL, which may be attributed to the inclusion of high-risk and T-lineage ALL cases in several primary studies, as these subtypes are often associated with more aggressive disease and potential variations in assay sensitivity and cutoff thresholds (30). The overlapping confidence intervals for AML, MDS, and ALL suggest that *WT1* expression is not specific to any single disease but instead reflects a shared molecular mechanism involved in malignant hematopoiesis. The consistently high prevalence across all three disorders underscores *WT1's* clinical significance as a leukemia biomarker, with potential utility in diagnosis, assessing disease burden, and monitoring treatment response (28).

The meta-analysis showed that *WT1* expression was 66.19% (95% CI 54.30–78.09%) in adults and 71.67% (95% CI 63.21–80.14%) in pediatric patients, with a prevalence of 74.26% (95% CI 63.83–84.88%) across both groups. These findings indicate that *WT1* overexpression is common in both adult and pediatric populations, demonstrating its consistent expression across age groups in acute leukemia and MDS. This reinforces its clinical utility as a biomarker for leukemia burden and MRD, supporting its role in diagnosis, disease monitoring, and treatment evaluation for patients of all ages (31).

This systematic review and meta-analysis showed that the prevalence of *WT1* expression was 71.13% (95% CI 63.72–78.54) in developing countries and 68.55% (95% CI 58.84–78.28) in developed countries. These findings indicate that *WT1* overexpression is consistently high across geographic regions, reinforcing its role as a leukemia biomarker regardless of country status (25). These differences may be explained by variations in study populations, disease presentation, and healthcare infrastructure, even though statistically insignificant differences were found. In developing countries, limited healthcare infrastructure may lead to delayed diagnosis, resulting in more advanced disease stages or more aggressive leukemic subtypes, which could contribute to higher *WT1* gene expression levels (32, 33).

The findings of this study indicated that the global pooled prevalence of *WT1* mutation in acute leukemia was 15.35% (95% CI 13.06–17.64). This finding indicates the mutation of the WT1 gene remains a moderate disease burden; this may be due to *WT1's* biological role in leukemogenesis and the heterogeneous landscape of acute leukemia. *WT1* mutations arise in a subset of leukemic clones, particularly in acute leukemia, and occur preferentially in specific molecular and cytogenetic subgroups, frequently coexisting with other oncogenic alterations and reflecting clonal evolution, resulting in the accumulation of leukemic cells and a moderate global burden (5).

The findings of this systematic review and meta-analysis indicated that the prevalence of *WT1* mutations was 18.91% in ALL (95% CI 9.39–28.44) and 14.59% in AML (95% CI 12.30–16.69). These results suggest that the prevalence of *WT1* mutations is comparable between ALL and AML. The overlapping confidence intervals likely reflect the absence of a significant difference in *WT1* mutation frequency between the two leukemia subtypes, highlighting *WT1* mutations as a common molecular event in both lymphoid and myeloid leukemogenesis. The wider confidence interval observed in ALL may reflect greater uncertainty, likely driven by smaller sample sizes, fewer included studies, and greater clinical and methodological heterogeneity. In contrast, the narrower confidence interval in AML suggests greater precision, probably due to larger study populations and more standardized molecular profiling in AML (34, 35).

The findings of this systematic review and meta-analysis showed that the prevalence of *WT1* mutations was 14.83% in adults (95% CI 12.41–17.24) and 21.83% in pediatric patients (95% CI 10.00–33.66). *WT1* mutations were, therefore, relatively common in both adult and pediatric leukemia patients, with a higher pooled prevalence observed in pediatric cases. The wider confidence intervals in pediatric patients likely result from reduced precision due to smaller sample sizes or variability in mutation–detection methods, rather than true age-related biological differences. Nevertheless, the overlapping confidence intervals indicate that *WT1* mutations are clinically relevant across age groups, reflecting shared molecular abnormalities. In contrast, the narrower confidence interval in adults suggests more precise estimates, likely due to larger study cohorts and more standardized molecular testing, particularly in adult AML (34, 36).

The findings of this systematic review and meta-analysis revealed that the prevalence of *WT1* mutation by country status was 20.18% in developing countries (95% CI 13.66–26.69) and 12.48% (95% CI 10.36–17.64) in developed countries. The findings showed a higher point estimate of WT1 mutation prevalence in developing settings; however, the overlapping confidence intervals between developing and developed countries suggest no statistically meaningful difference. This indicates that *WT1* mutations are reported in both contexts, and the relatively higher pooled prevalence observed in developing regions may reflect methodological variability rather than true epidemiological disparities. Factors such as small sample sizes, single-center study designs, limited eligible studies, and restricted access to advanced molecular diagnostic tools may contribute to this variation. In addition, differences in diagnostic capacity and healthcare infrastructure may lead to delayed presentation and a higher proportion of advanced or high-risk leukemia cases at diagnosis, potentially resulting in an apparent increase in *WT1*-mutated cases alongside wider confidence intervals and increased heterogeneity (34, 35, 37).

The findings of this systematic review and meta-analysis showed that the prevalence of WT1 mutation was 10.02% in studies conducted before 2016 (95% CI 8.45–11.58) and 21.05% in studies conducted during and after 2016 (95% CI 16.41–25.70). These results indicate a clear increase in the reported prevalence of *WT1* mutations over time, with a substantially higher pooled prevalence observed in studies published after 2016. This more than twofold rise may reflect advancements in methodological and technological approaches rather than a true shift in disease biology. Notably, improvements in molecular techniques, including the adoption of quantitative real-time PCR with standardized *WT1* transcript quantification and enhanced assay sensitivity, have likely enabled more accurate detection of low level or previously undetectable *WT1* mutations (24, 34). Greater standardization of *WT1* assessment in recent years has contributed to the higher reported prevalence by improving the detection rate of *WT1* mutations (38, 39).

In addition, changes in study design and study population selection may have contributed to the increased reporting of *WT1* gene mutations. Earlier studies often involved smaller, highly selected cohorts or focused on specific leukemia subtypes. In contrast, more recent studies are multicenter in nature, include larger and more heterogeneous patient populations, and assess *WT1* mutations longitudinally at diagnosis and during follow-up. These comprehensive methodologies are likely to yield higher pooled prevalence estimates; however, there is no biological evidence to suggest a true increase in *WT1* mutation frequency among leukemia patients (40, 41).

The findings of this systematic review and meta-analysis indicated that the prevalence of *WT1* mutations by region—the African region, the region of America, the European region, the EMRO (East Mediterranean region), the WPRO (Western Pacific region), and the SEARO (South East Asia Region)—was 23.45% (95% CI 6.47, 40.44), 13.50% (95% CI 9.49, 17.51), 13.52 (95% CI 9.25, 17.79), 7.15 (95% CI 4.63, 9.67), 17.82 (95% CI 9.63, 26.01), and 17.65 (95% CI 10.94, 24.35) with heterogeneity (I^2 = 89.2%, P = 0.000) (95% CI 10.94, 24.36), respectively. These findings demonstrate notable regional variation in the prevalence of *WT1* gene mutations worldwide. The highest pooled prevalence was observed in Africa, followed by the SEARO and WPRO regions. The relatively higher prevalence reported in Africa may be attributed to differences in genetic background, leukemia subtype distribution, environmental exposures, diagnostic capacity, and study sample sizes, as reflected by the comparatively wide confidence interval (42, 43).

The SEAPRO and WPRO regions demonstrated moderately high prevalence, which may be influenced by improved molecular diagnostic techniques, higher study density, or population-specific genetic susceptibility. In contrast, the Region of the Americas and Europe showed relatively lower and comparable magnitudes, whereas the lowest pooled prevalence was reported in EMRO. The lower prevalence reported in EMRO could be attributable to limited availability of molecular testing, fewer published studies, or under-representation in pooled analyses (44, 45).

Strengths and limitations of the study

Strengths

To our knowledge, this is the first systematic study and meta-analysis to assess the pooled prevalence of *WT1* gene expression and mutation among patients with hematological malignancies (acute leukemia and MDS) worldwide and how it varies by age difference, country status, type of hematological malignancy (ALL, AML, and MDS), and year of publication. This comprehensive review and meta-analysis provides solid evidence of pooled prevalence.

Limitations

This study has several limitations. First, the analysis did not assess determinant factors associated with *WT1* expression or mutation. Although *WT1* detection in all included studies was performed using quantitative PCR (qPCR), variations in laboratory protocols, assay sensitivity, and the absence of standardized cutoff values for defining *WT1* overexpression may have affected the comparability of results and contributed to heterogeneity across studies. Inconsistent thresholds for *WT1* positivity may also have introduced differences in outcome classification, potentially reducing the precision of the pooled estimates of this study. In addition, variability in reporting, language restrictions, and the lack of patient-level data limited the ability to fully synthesize WT1-related outcomes. Publication bias cannot be completely excluded, and the uneven geographic distribution of studies, with limited data from some regions, may affect the global generalizability of the findings. The evidence certainty was low to very low based on the GRADE approach due to observational designs and high heterogeneity. Thus, findings should be interpreted cautiously and require further robust studies.

Conclusion

This comprehensive systematic review and meta-analysis has shown that the global prevalence of *WT1* gene expression among patients with leukemia and MDS was 69.74%. This indicates that *WT1* expression is highly prevalent in patients with acute leukemia and MDS. *WT1* expression was consistently high across leukemia subtypes (AML, ALL, and MDS), age groups, geographic locations, and country income levels, indicating its robustness as a leukemic biomarker for malignant hematopoiesis and disease burden.

In contrast, the *WT1* mutation in acute leukemia was 15.35%, with moderate global prevalence, comparable frequencies across leukemia types and age categories, occurring in a fraction of leukemia patients and with comparable frequencies across leukemia subtypes and age groups. The observed heterogeneity and temporal, regional, and socioeconomic variations in *WT1* mutation prevalence are most likely attributable to differences in diagnostic capacity and advances in molecular testing rather than true biological differences. Overall, these findings highlight the central role of *WT1* in leukemia biology and its relevance to disease characterization and monitoring. The overall finding the evidence was rated as low certainty due to very high heterogeneity and reliance on observational studies; therefore, the findings should be interpreted with caution.

Recommendation

In patients with acute leukemia and myelodysplastic syndromes, we suggest using *WT1* gene expression as a molecular biomarker for prognostic stratification and disease burden evaluation, particularly in settings with access to standardized molecular assays. We suggest that *WT1* expression should be routinely assessed in clinical practice to aid in diagnosis, burden, and MRD monitoring in patients with acute leukemia and MDS. We suggest selective testing for *WT1* mutations in defined clinical or research contexts, particularly in high-risk disease subsets, rather than universal screening.

Standardized and sensitive molecular methods, particularly quantitative real-time PCR with consistent cutoff values, shall be used to promote comparability across studies and settings. Future research is recommended to concentrate on large, multicenter, longitudinal studies, particularly in under-represented areas, to reduce heterogeneity and refine prevalence estimates. In addition, we recommend advanced molecular and large population-based research to better understand how *WT1* mutations interact with other molecular abnormalities and how they affect therapy responsiveness and survival outcomes.

Clinical trial number: Not applicable, as this study is a systematic review and meta-analysis.

Ethics approval and consent to participate: N/A

Consent for publication: N/A

Availability of data and materials: All data generated or analyzed during this study are included in this published article and its accompanying supplementary information files.

Declaration: The authors declare that there is no competing interest among them.

Funding: N/A

Abbreviations: *WT1:* Wilms Tumor 1 gene; MDS: Myelodysplastic syndromes, CML: Chronic Myeloid Leukemia, MRD: Minimal Residual Disease, AML: Acute Myeloid Leukemia, PRISMA-P: Preferred Reporting Items for Systematic Review and Meta-analysis Protocols, MEDLINE: Medical Literature Analysis and Retrieval System Online, HINARI: Health Internetwork Access to Research Initiative, CINAHL: Cumulative Index to Nursing and Allied Health Literature, SID: Scientific Information Database, JBI: Joanna Briggs Institute, USA: United States of America, UK: United Kingdom, CI: Confidence interval, WHO: World Health Organization. EMRO: East Mediterranean region, WPRO: Western Pacific region, SEA RO: South East Asia region

Author's contribution: SS conceived and designed the study. LGB, SS, and GM participated in article search, and data extraction. SA, TW, DF, LGB, SS, and GM conducted a quality assessment of the included studies and performed the statistical analysis and interpretation of the data. LGB drafted the manuscript. SS, TW and GM checked the validity and monitored the overall process.

LGB, GM, SS, and TW critically reviewed the manuscript. All the authors read and approved the final manuscript.

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